Employee Success Analytics at NextGen Corp.

Query and Data Outputs

A. Understanding the employee turnover trends and identifying the root causes of high turnover rates.

1. Who are the top 5 highest serving employees?

```
select
    e.first_name|| ' ' || e.last_name as "Employee Name",
    e.hire_date as "Start Date",
    coalesce(t.turnover_date, current_date) as "End Date",
    (coalesce(t.turnover_date, current_date) - e.hire_date)/365  as "Years of Service"
from employee e
left join turnover t
    on e.employee_id = t.employee_id
order by "Years of Service" desc
limit 5;
```

	Employee Name text	Start Date date	End Date date	Years of Service integer
1	David Moore	2015-06-30	2025-06-29	10
2	John Johnson	2015-10-27	2025-06-29	9
3	Frank Smith	2015-09-08	2025-04-07	9
4	Frank Johnson	2016-06-12	2025-06-29	9
5	John Doe	2016-02-16	2024-11-24	8

2. What is the turnover rate for each department?

```
select
    d.department_name as "Department",
    count(e.employee_id) as "Employees",
    count(t.employee_id) as "Turnover",
    round(count(t.employee_id)*1.0/count(e.employee_id)*100,2) as "Turnover Rate"
from department d
join employee e
    on d.department_id = e.department_id
left join turnover t
    on t.employee_id = e.employee_id
group by d.department_name;
```

	Department character varying (30)	Employees bigint	Turnover bigint	Turnover Rate numeric
1	Engineering	6	4	66.67
2	Sales	29	8	27.59
3	HR	11	3	27.27
4	Marketing	14	13	92.86

3. Which employees are at risk of leaving based on their performance?

(a) Employees with Lowest 10 Performance Score

```
select
    e.first_name|| ' ' || e.last_name as "Employee Name",
    p.performance_score as "Performance Score"
from employee e
inner join performance p
    on e.employee_id = p.employee_id
order by "Performance Score" asc
limit 10;
```

	Employee Name text	Performance Score numeric (2,1)
1	Frank Lee	3.0
2	Eve Davis	3.0
3	Grace Wilson	3.0
4	Charlie Smith	3.0
5	David Moore	3.0
6	Frank Wilson	3.0
7	Grace Smith	3.0
8	John Brown	3.0
9	Jane Wilson	3.1
10	Jane Wilson	3.1

(b) Employees with Lowest 10 Average Performance Score

```
select
    e.first_name|| ' ' || e.last_name as "Employee Name",
    round(avg(p.performance_score), 2) as "Average Performance Score"
from employee e
inner join performance p
    on e.employee_id = p.employee_id
group by e.first_name|| ' ' || e.last_name
order by "Average Performance Score" asc
limit 10;
```

	Employee Name text	Average Performance Score numeric
1	Grace Wilson	3.43
2	Jane Wilson	3.58
3	Eve Davis	3.65
4	David Smith	3.72
5	Charlie Lee	3.77
6	Alice Wilson	3.77
7	Alice Brown	3.80
8	John Davis	3.83
9	Grace Smith	3.83
10	Bob Green	3.85

4. What are the main reasons employees are leaving the company?

```
select
    t.reason_for_leaving as "Reason for Leaving",
    count(t.reason_for_leaving) as "Count of Reason"
from turnover t
group by "Reason for Leaving"
order by "Count of Reason" desc;
```

	Reason for Leaving text	Count of Reason bigint
1	Personal	11
2	Found Another Job	7
3	Career Growth	5
4	Retired	5

B. Evaluating employee performance across different departments and identifying areas where performance can be improved.

5. How many employees has left the company? (a) Company-wide

```
select
    count(t.reason_for_leaving) as "Total Number who left the company"
from  turnover t;
```

	Total Number who left the company bigint
1	28

(b) By Department

```
select d.department_name as "Department",
    count(t.reason_for_leaving) as "Number who left the company"
from  department d
left join turnover t
    on d.department_id =  t.department_id
group by "Department";
```

	Department character varying (30)	Number who left the company bigint
1	Engineering	9
2	Sales	4
3	HR	6
4	Marketing	9

6a. How many employees have a performance score of 5.0?

```
select
    performance_score as "Performance Score",
    count(performance_score) as "Count of Score"
from performance
where performance_score = 5.0
group by performance_score;
```

	Performance Score numeric (2,1)	Count of Score bigint
1	5.0	9

6b. How many employees have a performance score of below 3.5? (i) Company-wide

```
select
    count(performance_score) as "Total Count of Score"
from performance
where performance_score < 3.5;
```

	Total Count of Score bigint
1	71

(ii) By Low Performance Score

```
select
    count(performance_score) as "Total Count of Score"
from performance
where performance_score < 3.5;
```

	Performance Score numeric (2,1)	Count of Score bigint
1	3.3	15
2	3.4	13
3	3.0	8
4	3.2	19
5	3.1	16

7a. Which department has the most employees with a performance score of 5.0?

```
select d.department_name as "Department", p.performance_score as "Performance Score",
    count(p.employee_id) as "Number of Employees"
from department d
left join performance p
    on d.department_id = p.department_id
where p.performance_score = 5.0
group by d.department_name, p.performance_score
order by "Number of Employees" desc;
```

	Department character varying (30)	Performance Score numeric (2,1)	Number of Employees bigint
1	Marketing	5.0	5
2	Engineering	5.0	3
3	Sales	5.0	1

7b. Which department has the most employees with a performance score below 3.5?

```
select d.department_name as "Department",
    count(p.performance_score) as "Number of Employee with Score less than 3.5"
from performance p
join department d
    on d.department_id = p.department_id
where p.performance_score < 3.5
group by "Department"
order by "Number of Employee with Score less than 3.5"  desc;
```

	Department character varying (30)	Number of Employee with Score less than 3.5 bigint
1	Marketing	28
2	Engineering	22
3	Sales	12
4	HR	9

8. What is the average performance score by department?

```
select
    d.department_name as "Department",
    round(avg(p.performance_score), 2) as "Average Performance Score"
from performance p
join department d
    on d.department_id = p.department_id
group by d.department_name
order  by "Average Performance Score" desc;
```

	Department character varying (30)	Average Performance Score numeric
1	Marketing	4.13
2	Engineering	4.10
3	HR	4.05
4	Sales	4.00

C. Analyze salary distribution and ensure fair compensation based on performance and departmental benchmarks.

9. What is the total salary expense for the company?

```
select
    sum(salary.salary_amount) as "Total Salary Expense"
from salary;
```

	Total Salary Expense bigint
1	4850000

10. What is the average salary by job title?

```
select e.job_title as "Job Title", round(avg(s.salary_amount),  2)  as "Average Salary"
from employee e
left join salary s
    on e.employee_id = s.employee_id
group by e.job_title
order  by "Average Salary" asc;
```

	Job Title character varying (30)	Average Salary numeric
1	Marketing Specialist	77857.14
2	Engineer	80000.00
3	Sales Manager	80000.00
4	HR Specialist	81818.18
5	Sales Representative	84285.71

11. How many employees earn above 80,000? (a) Company-wide

```
select count(s.employee_id) as "Number of Employee Earning above 80,000"
from salary s
where s.salary_amount > 80000;
```

	Number of Employee Earning above 80,000 bigint
1	26

(b) By Department

```sql
select d.department_name as "Department",
    count(s.employee_id) as "Number of Employee Earning above 80,000"
from  department d
left join salary s
    on d.department_id =  s.depaartment_id
where s.salary_amount > 80000
group by "Department";
```

	Department character varying (30)	Number of Employee Earning above 80,000 bigint
1	Sales	4
2	Engineering	8
3	HR	5
4	Marketing	9

(c) By Job Title

```sql
select e.job_title as "Job Title",
    count(s.employee_id) as "Number of Employee Earning above 80,000"
from  employee e
left join salary s
    on e.employee_id =  s.employee_id
where s.salary_amount > 80000
group by "Job Title";
```

	Job Title character varying (30)	Number of Employee Earning above 80,000 bigint
1	HR Specialist	6
2	Sales Representative	7
3	Marketing Specialist	4
4	Engineer	2
5	Sales Manager	7

12. How does performance correlate with salary across departments?
(a) For the lowest Salary Amount of 60,000 by Employee Name.

```
select e.first_name|| ' ' || e.last_name as "Employee Name",
    round(avg(p.performance_score), 2) as "Average Performance Score",
    s.salary_amount as "Salary Amount"
from employee e
join performance p on  e.employee_id = p.employee_id
left join salary s on  p.employee_id = s.employee_id
where s.salary_amount = 60000
group by p.employee_id, "Employee Name", s.salary_amount
order by "Average Performance Score";
```

	Employee Name text	Average Performance Score numeric	Salary Amount integer
1	Jane Wilson	3.58	60000
2	Bob Green	3.85	60000
3	David Lee	3.85	60000
4	David Moore	3.90	60000
5	John Wilson	3.93	60000
6	Jane Moore	4.00	60000
7	David Wilson	4.03	60000
8	David Green	4.08	60000
9	Frank Smith	4.18	60000
10	Frank Moore	4.25	60000
11	Bob Moore	4.27	60000
12	John Johnson	4.50	60000

(b) For the highest Salary Amount of 100,000 by Employee Name.

```sql
select e.first_name|| ' ' || e.last_name as "Employee Name",
    round(avg(p.performance_score), 2) as "Average Performance Score",
    s.salary_amount as "Salary Amount"
from employee e
join performance p on  e.employee_id = p.employee_id
left join salary s on  p.employee_id = s.employee_id
where s.salary_amount = 100000
group by p.employee_id, "Employee Name", s.salary_amount
order by "Average Performance Score";
```

	Employee Name text	Average Performance Score numeric	Salary Amount integer
1	John Davis	3.83	100000
2	Frank Green	3.90	100000
3	Charlie Wilson	3.92	100000
4	Frank Wilson	3.92	100000
5	Alice Miller	4.02	100000
6	Eve Wilson	4.05	100000
7	Hannah Doe	4.13	100000
8	Jane Green	4.20	100000
9	Frank Johnson	4.23	100000
10	Jane Lee	4.32	100000
11	Charlie Moore	4.33	100000
12	Bob Doe	4.40	100000
13	John Green	4.43	100000
14	Alice Doe	4.52	100000

(c) For the lowest Salary Amount of 60,000 by Department.

```sql
select d.department_name as "Department",
    round(avg(p.performance_score), 2) as "Average Performance Score",
    s.salary_amount as "Salary Amount"
from department d
join performance p on  d.department_id = p.department_id
left join salary s on  p.employee_id = s.employee_id
where s.salary_amount = 60000
group by p.employee_id, "Department", s.salary_amount
order by "Average Performance Score";
```

	Department character varying (30)	Average Performance Score numeric	Salary Amount integer
1	HR	3.58	60000
2	Marketing	3.85	60000
3	Engineering	3.85	60000
4	Marketing	3.90	60000
5	Marketing	3.93	60000
6	Marketing	4.00	60000
7	Marketing	4.03	60000
8	Engineering	4.08	60000
9	Engineering	4.18	60000
10	Marketing	4.25	60000
11	Engineering	4.27	60000
12	Engineering	4.50	60000

(c) For the highest Salary Amount of 100,000 by Department.

```sql
select d.department_name as "Department",
    round(avg(p.performance_score), 2) as "Average Performance Score",
    s.salary_amount as "Salary Amount"
from department d
join performance p on  d.department_id = p.department_id
left join salary s on  p.employee_id = s.employee_id
where s.salary_amount = 100000
group by p.employee_id, "Department", s.salary_amount
order by "Average Performance Score";
```

	Department character varying (30)	Average Performance Score numeric	Salary Amount integer
1	Marketing	3.83	100000
2	Marketing	3.90	100000
3	Marketing	3.92	100000
4	Engineering	3.92	100000
5	Engineering	4.02	100000
6	HR	4.05	100000
7	HR	4.13	100000
8	Marketing	4.20	100000
9	Engineering	4.23	100000
10	Sales	4.32	100000
11	Engineering	4.33	100000
12	Marketing	4.40	100000
13	Marketing	4.43	100000
14	Engineering	4.52	100000